



PRECUSOR ANALYSIS REPORT: LOCKERGOGA RANSOMWARE ATTACK ON NORSK HYDRO 2019

Cybersecurity for the Operational Technology
Environment (CyOTE)

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PRECUSOR ANALYSIS REPORT: LOCKERGOGA RANSOMWARE ATTACK ON NORSK HYDRO 2019

1. EXECUTIVE SUMMARY

The LockerGoga Ransomware Attack on Norsk Hydro 2019 Precursor Analysis Report leverages publicly available information about the Norsk Hydro cyber attack and catalogs anomalous observables for each technique employed in the attack. This analysis is based upon the methodology of the Cybersecurity for the Operational Technology Environment (CyOTE) program.

Norsk Hydro is one of the largest energy producers in Norway, powering over 900,000 homes per year. The company is also one of the largest aluminum manufacturers in the world.¹ On 18 March 2019, adversaries used a malware known as LockerGoga to carry out an attack on the company. The attack began when adversaries used a spoofed customer account to engage a Norsk Hydro employee, deceiving the victim into downloading a malicious attachment through a spearphishing email.

LockerGoga propagated across the victim network, causing Norsk Hydro to shut down over 23,000 PCs and 3,000 servers.² The company's 35,000 employees worked manually for approximately three weeks following the attack to complete tasks typically facilitated by both Information Technology (IT) and Operational Technology (OT) systems.³ Norsk Hydro began to recover approximately four weeks after the attack, which cost the company over \$71 million in losses.⁴ Norsk Hydro decided not to pay the ransom demanded by the adversaries and instead planned to recover data through cloud backups, with full recovery taking over a year. The financial loss and impact on operations caused the company to implement significant changes to its cybersecurity processes and operational practices.

Researchers and analysts identified 13 unique techniques during the attack with a total of 157 observables using the MITRE ATT&CK[®] for Industrial Control Systems framework. The CyOTE program assesses observables accompanying techniques used prior to the triggering event to identify opportunities to detect malicious activity. If observables accompanying the attack techniques are perceived and investigated prior to the triggering event, earlier comprehension of malicious activity can take place. Eight of the identified techniques used during the Norsk Hydro cyber attack were precursors to the triggering event. Case study analysis identified 123 observables associated with these precursor techniques, 105 of which were assessed to have an increased likelihood of being perceived in the 30 days preceding the triggering event. The response and comprehension time could have been reduced if the observables had been identified earlier.

The information gathered in this report contributes to a library of observables tied to a repository of artifacts, data sources, and technique detection references for practitioners and developers to support the comprehension of indicators of attack. Asset owners and operators can use these products if they experience similar observables or to prepare for comparable scenarios.

2. INTRODUCTION

The [Cybersecurity for the Operational Technology Environment \(CyOTE\)](#) program developed capabilities for energy sector organizations to independently identify adversarial tactics and techniques within their operational technology (OT) environments. Led by Idaho National Laboratory (INL) under leadership of the Department of Energy (DOE) Office of Cybersecurity, Energy Security, and Emergency Response (CESER), CyOTE is a partnership with energy sector owners and operators whose goal is to tie the impacts of a cyber attack to anomalies in the OT environment to determine whether the anomalies have a malicious cyber cause.

2.1. APPLYING THE CYOTE METHODOLOGY

The CyOTE methodology, as shown in Figure 1. CyOTE Methodology, applies fundamental concepts of perception and comprehension to a universe of knowns and unknowns increasingly disaggregated into observables, anomalies, and triggering events. The program utilizes MITRE's ATT&CK® Framework for Industrial Control Systems (ICS) as a common lexicon to assess triggering events. By leveraging the CyOTE methodology with existing commercial monitoring capabilities and manual data collection, energy sector partners can understand relationships between multiple observables, which could represent a faint signal of an attack requiring investigation. CyOTE can assist organizations in prioritizing their OT environment visibility investments.

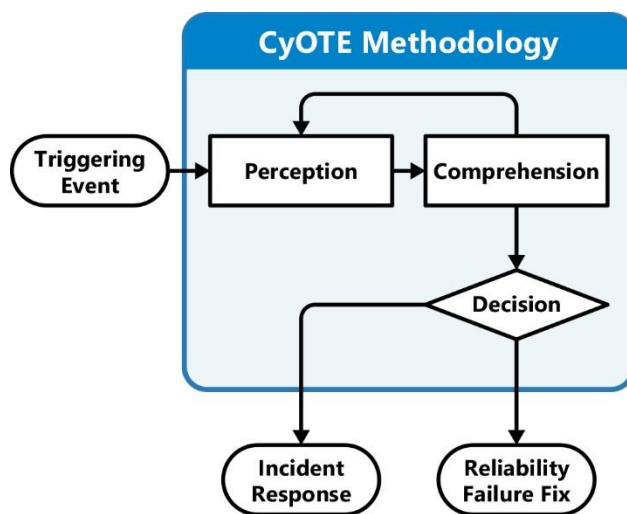


Figure 1. CyOTE Methodology

Historical case studies such as this one support continued learning through analysis of incidents that have impacted OT. This precursor analysis report is based on publicly available reports and provides examples of how key concepts in the CyOTE methodology appear in the real world, providing insights on how similar novel attacks could be detected earlier and therefore mitigated. The analysis enables OT personnel to independently identify observables associated with techniques known to be indicators of attack within OT environments. The identified observables highlight anomalous events for further investigation, which could enhance comprehension of malicious activity.

A timeline of events based on the CyOTE methodology portrays the attack-related observables associated with the case study's cyber attack. The timeline includes assessed dates, the triggering event, and comprehension of malicious activity by the organization. The point on this timeline when each technique appears is critical to the organization's ability to perceive and comprehend the associated malicious activity. Perception of techniques early in the timeline is critical, since halting those techniques will generally have greater potential to limit additional attack vectors using other techniques, defeat the cyber attack, and limit damage to operations.

Each technique has an assessed perceivability. Perceivability is a function of the number of observables and the potential for personnel to detect those observables. If a technique includes effects which personnel may detect, such as deletion or modification of system files or required user execution, then the technique would be more perceivable.

Differences in infrastructure and system configurations may present different challenges and opportunities for observable detection. For example, architecture-wide endpoint monitoring is likely to improve the perceivability of techniques which modify host files, such as the Data Destruction technique (T0809) for Inhibit Response Function and Theft of Operational Information technique (T0882) for Impact. Network monitoring and log analysis capabilities are likely to improve perceivability of techniques which create malicious network traffic, such as the Standard Application Layer Protocol technique (T0869) for Command and Control, External Remote Services technique (T0822) for Initial Access, and Connection Proxy technique (T0884) for Command and Control. Alternatively, enhancing the monitoring parameters of system files would increase the perceivability of techniques such as Data from Information Repositories technique (T0811) for Collection and the Service Stop technique (T0881) for Inhibit Response Function.

Comprehension can be further enhanced by technique artifacts created when adversaries employ certain attack techniques. The CyOTE program provides organizations with a [library of observables](#) reported in each historical case. The library can be used in conjunction with a repository of artifacts, data sources, and technique detection references for practitioners and developers to support the comprehension of indicators of attack.

2.2. BACKGROUND ON THE ATTACK

Adversaries gained initial access to the Norsk Hydro network through a spearphishing email three months (D-90) before the Service Stop (D-0) triggering event on 18 March 2019.⁵ The spearphishing email attachment, with embedded LockerGoga malware, provided the adversaries with initial access to the company's network.

LockerGoga propagated across Norsk Hydro's network, encrypting files and data. An operator working at the extrusion plant was the first to notice a ransom note on a desktop machine at 10:45 PM^a on 18 March (D-0).⁶ The extrusion plant experienced the largest financial losses among the Norsk Hydro subsidiaries operating across 50 countries.⁷

At about 3:00 AM on 19 March (H+5), Norsk Hydro's Head of Enterprise Service Management made the decision to shut down over 23,000 PCs and 3,000 servers.⁸

CyOTE analysts assess that Norsk Hydro achieved comprehension of the attack on 19 March (D+1), when company officials held a press conference and announced they would not pay the ransom.⁹

A timeline of adversarial techniques is shown in Figure 2. The timeline includes the estimated number of days prior to and after the triggering event. The timeline after the triggering event includes the assessed victim comprehension timeline.

Norsk Hydro's 35,000 employees carried out both Information Technology (IT) and OT tasks manually for nearly a month (D+30) after the attack and were unable to fully recover all business areas until over a year later (D+365).¹⁰ The attack on Norsk Hydro cost the company over \$71 million in losses.¹¹

Analysis identified 13 unique techniques in a sequence and timeframe likely used by adversaries during this cyber attack ([Error! Reference source not found. 1](#)). These attack techniques are defined according to MITRE's ATT&CK® for ICS framework.

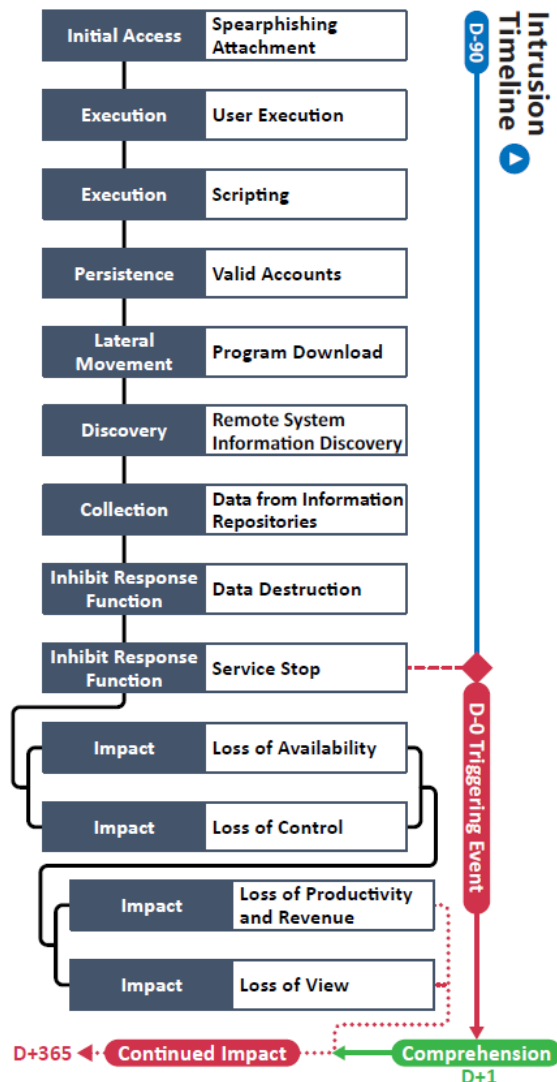


Figure 2. Intrusion Timeline

^a All times given are for GMT/UTC.

Table 1. Techniques Used in the Norsk Hydro Cyber Attack

Initial Access	Execution	Persistence	Privilege Escalation	Evasion	Discovery	Lateral Movement	Collection	Command and Control	Inhibit Response Function	Impair Process Control	Impact
Data Historian Compromise	Change Operating Mode	Modify Program	Exploitation for Privilege Escalation	Change Operating Mode	Network Connection Enumeration	Default Credentials	Automated Collection	Commonly Used Port	Activate Firmware Update Mode	Brute Force I/O	Damage to Property
Drive-by Compromise	Command-Line Interface	Module Firmware	Hooking	Exploitation for Evasion	Network Sniffing	Exploitation of Remote Services	Data from Information Repositories	Connection Proxy	Alarm Suppression	Modify Parameter	Denial of Control
Engineering Workstation Compromise	Execution through API	Project File Infection		Indicator Removal on Host	Remote System Discovery	Lateral Tool Transfer	Default Operating Mode	Standard Application Layer Protocol	Block Command Message	Module Firmware	Denial of View
Exploit Public-Facing Application	Graphical User Interface	System Firmware		Masquerading	Remote System Information Discovery	Program Download	I/O Image		Block Reporting Message	Spoof Reporting Message	Loss of Availability
Exploitation of Remote Services	Hooking	Valid Accounts		Rootkit	Wireless Sniffing	Remote Services	Man in the Middle		Block Serial COM	Unauthorized Command Message	Loss of Control
External Remote Services	Modify Controller Tasking			Spoof Reporting Message		Valid Accounts	Monitor Process State		Data Destruction		Loss of Productivity and Revenue
Internet Accessible Device	Native API						Point & Tag Identification		Denial of Service		Loss of Protection
Remote Services	Scripting						Program Upload		Device Restart/Shutdown		Loss of Safety
Replication Through Removable Media	User Execution						Screen Capture		Manipulate I/O Image		Loss of View
Rogue Master							Wireless Sniffing		Modify Alarm Settings		Manipulation of Control
Spearphishing Attachment									Rootkit		Manipulation of View
Supply Chain Compromise									Service Stop		Theft of Operational Information
Wireless Compromise									System Firmware		

Table 2. Precursor Analysis Report Quantitative Summary

Precursor Analysis Report Quantitative Summary	Totals
MITRE ATT&CK® for ICS Techniques	13
Technique Observables	157
Precursor Techniques	8
Precursor Technique Observables	123
Highly Perceivable Precursor Technique Observable	105

3. OBSERVABLE AND TECHNIQUE ANALYSIS

The following analysis may assist organizations in identifying malicious cyber activity earlier and more effectively. The following techniques and observables were compiled from publicly available sources and correlated with expert analysis.

3.1. SPEARPHISHING ATTACHMENT TECHNIQUE (T0865) FOR INITIAL ACCESS

Approximately three months before the attack, adversaries sent a spearphishing attachment to a Norsk Hydro employee from a spoofed customer account, mirroring a legitimate employee conversation with customers.¹² The attachment contained LockerGoga, a trojan malware.

IT Staff, IT Cybersecurity, OT Staff, OT Cybersecurity, Engineering, Support Staff, and Management personnel may have been able to observe the spearphishing email.

A total of seven observables were identified with the use of the [Spearphishing Attachment technique \(T0865\)](#). This technique is important for investigation because it allowed adversaries to gain initial access. This technique appears first in the timeline and responding to it will effectively halt all future events. Terminating the chain of techniques at this point would prevent adversaries from gaining access to the system.

Of the seven observables associated with this technique, six are assessed to be highly perceivable (Anomalous Email with Attachment; Email Attachment with Anomalous Embedded Scripts; Email with Anomalous Word Document; Email with Anomalous Word Document with Object Linking and Embedding (OLE); Email with Anomalous Excel File; Email with Anomalous Excel File with Object Linking and Embedding (OLE)).

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 29 artifacts could be generated by the Spearphishing Attachment technique
Technique Observers^b	IT Staff, IT Cybersecurity, OT Staff, OT Cybersecurity, Engineering, Support Staff, and Management
Resources	Technique Detection References

^b Observer titles are adapted from the Job Role Groupings listed in [the SANS ICS Job Role to Competency Level Poster](#). CyOTE products utilize these job categories rather than organizational titles to both support comprehensive analysis and preserve anonymity within the victim organization. A complete list of potential observers can be found in [Appendix C](#).

3.2. USER EXECUTION TECHNIQUE (T0863) FOR EXECUTION

A Norsk Hydro employee likely opened the spearphishing email and unknowingly executed the infected attachment two to three weeks prior to the triggering event (18 March 2019), granting adversaries access to the network.¹³

LockerGoga initially downloads malicious executables into the %TEMP% folder on the host machine. The malware then creates a file in the %TEMP% folder with the following schema: %TEMP%\svc{random}.{random number}.exe. LockerGoga also creates another file in the %TEMP% folder with the following schema: File e%TEMP%\tgytutrc{4 Random Numbers}.exe. An anomalous process is spawned from a binary renamed from PsExec with the -m parameter. LockerGoga then executes both folders, which spawn up a primary and several secondary processes.¹⁴ The primary process leverages the Boost InterProcess Communication (IPC) Library to communicate with the secondary processes through writes to shared memory space. The primary process searches for files to encrypt, passing the paths of these selected files to the secondary processes through the shared memory space via the IPC method. The secondary processes take these paths and perform the encryption on each file path passed by the primary process.

IT Staff, IT Cybersecurity, and OT Cybersecurity personnel may have been able to observe the LockerGoga malware, email attachment, and associated processes.

A total of 13 observables were identified with the use of the [User Execution technique \(T0863\)](#). This technique is important for investigation because it allows the malware access to the host. This technique will generate anomalous files and directories on the host. This technique modifies the host operating system files via the download and installation of the LockerGoga malware, placing the host into a modified or compromised state. This technique also appears relatively early in the timeline and responding to it would effectively halt the adversaries' lateral movement. Terminating the chain of techniques at this point would prevent the malware from infecting the host, limiting operational damage in both the IT and OT environments.

Of the 13 observables associated with this technique, eight are assessed to be highly perceivable (Anomalous Access from External Host; Anomalous Remote Process Execution Using Renamed PsExec; Creation of Anomalous Executable Files in %TEMP%\svc{random}.{random number}.exe; Anomalous File Execution from %TEMP%\svc{random}.{random number}.exe-{random} -{random} {random}; Anomalous File Execution from %TEMP%\tgytutrc{4 Random Numbers}.exe; Anomalous Process Spawned from PsExec with the -m Parameter; Anomalous Process Spawned from Anomalous Binary Renamed from PsExec with the -m Parameter; A New Process Has Been Created Windows Event Log (Windows Event ID 4688)).

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 23 artifacts could be generated by the User Execution technique
Technique Observers	IT Staff, IT Cybersecurity, OT Cybersecurity
Resources	Technique Detection References

3.3. SCRIPTING TECHNIQUE (T0853) FOR EXECUTION

Once installed, LockerGoga can modify the user's credentials using Mimikatz, Cobalt Strike, PsExec, and Metasploit, allowing adversaries to access privileged accounts within the network.^{15,16} Malware analysis revealed that a renamed PsExec tool was dropped and executed, generating logins into the "usual subject" Windows Event Log and pulling administrative rights within Windows Active Directory (AD).¹⁷ CyOTE analysis has determined successful and failed login attempts in the Windows AD likely could have been observed within the Norsk Hydro network displaying Event ID 4624 if login was successful or if login failure occurred.

PsExec displays two Event IDs, 4697 for a new service installation and 7045 for a new service creation on the local machine.¹⁸ CyOTE analysis determined that adversaries likely attempted to change the target user account's password. Event ID 4723 is triggered when a user attempts to change their own user password and displays it in the Windows AD. Each time adversaries encrypt a file with LockerGoga the Current User HKEY is modified. LockerGoga leaves a ransom note in a file named README_LOCKED.txt within the desktop folder.¹⁹ In the case of Norsk Hydro, employees found this ransom note on an infected desktop workstation.²⁰

IT Staff, IT Cybersecurity, and Support Staff personnel may have been able to observe the tools, Windows Event IDs, and commands associated with LockerGoga.

A total of 25 observables were identified with the use of the [Scripting technique \(T0853\)](#). This technique is important for investigation because the system event logs in the AD could have potentially notified observers of the Norsk Hydro network intrusion. This technique also modifies the host operating system files via the creation of anomalous services and modification of user accounts, resulting in the host being placed into a modified or compromised state. This technique appears early in the attack timeline and responding to it will disrupt further infection of a network. Terminating the chain of techniques at this point would limit adversaries' ability to cause operational damage with LockerGoga malware.

Of the 25 observables associated with this technique, 17 are assessed to be highly perceivable as indicated in [Appendix A](#).^c

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 12 artifacts could be generated by the Scripting technique
Technique Observers	IT Staff, IT Cybersecurity, Support Staff
Resources	Technique Detection References

^c For purposes of clarity, excessively long lists of highly perceivable observables will be listed only in the associated technique tables in Appendix A, rather than in the technique narrative sections.

3.4. VALID ACCOUNTS TECHNIQUE (T0859) FOR PERSISTENCE

The LockerGoga malware compromised standard user accounts within the Norsk Hydro network, eventually gaining privileges to control the entire IT infrastructure.^{21,22}

IT Staff and IT Cybersecurity personnel may have been able to observe anomalous traffic between Norsk Hydro's systems and unknown external IP addresses.

A total of five observables were identified with the use of the [Valid Accounts technique \(T0859\)](#). This technique is important for investigation because compromised credentials may be used to bypass access controls to various resources within a network or grant an adversary increased privileges to specific systems and devices. This technique appears relatively early in the timeline and responding to it will limit lateral movement and access to protected systems. Terminating the chain of techniques at this point would prohibit adversaries from gaining unauthorized access to the network.

Of the five observables associated with this technique, three are assessed to be highly perceivable (Anomalous Interactive Successful Logons (Windows Event ID 4624 Type 3 or 10); An Account Was Successfully Logged on Windows Log Event (Windows Event ID 4624); Special Privileges Assigned to New Logon Windows Log Event (Windows Event ID 4672)).

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 16 artifacts could be generated by the Valid Accounts technique
Technique Observers	IT Staff, IT Cybersecurity
Resources	Technique Detection References

3.5. PROGRAM DOWNLOAD TECHNIQUE (T0843) FOR LATERAL MOVEMENT

Adversaries can install LockerGoga by equipping the malicious email attachment with an executable capable of downloading programs. In the case of the attack on Norsk Hydro, LockerGoga's payload likely sat dormant on an infected computer on the company's network for two to three weeks before the first ransom note appeared on the night of 18 March.²³

LockerGoga downloads different processes to perform and accelerate file encryption within the system.²⁴ In addition, adversaries downloaded Metasploit and Cobalt Strike to pivot to other systems within Norsk Hydro's network and used Mimikatz to gather user credentials.²⁵

LockerGoga malware demonstrates unusual Central Processing Unit (CPU) behavior due to its ability to utilize all available CPU resources.²⁶ LockerGoga's main access point is Windows AD domain services through the Domain Controller.²⁷

IT Staff, IT Cybersecurity, and OT Cybersecurity personnel may have been able to observe the downloading of Metasploit, Mimikatz, and Cobalt Strike. Personnel may have also observed unusual CPU activity.

A total of 18 observables were identified with the use of the [Program Download technique \(T0843\)](#). This technique is important for investigation because it enables LockerGoga to perform file encryption, concealing content information with code. Through downloading of various modules, LockerGoga is able to perform file encryption. This technique modifies the host operating system files via the creation of anomalous services and modification of user accounts, placing the host into a modified or compromised state. This technique appears in the middle of the timeline and responding to it will prevent the installation of additional malware. Terminating the chain of techniques at this point would prevent the compromise of host devices.

Of the 18 observables associated with this technique, 9 are assessed to be highly perceivable (Anomalous Application Event Log Entries; Application Event Log Entries Associated with Anomalous Executable; Network Traffic Associated with Anomalous Metasploit Content; Network Traffic Associated with Metasploit Download; Network Traffic Associated with Anomalous Mimikatz Content; Network Traffic Associated with Mimikatz Download; Network Traffic Associated with Anomalous Cobalt Strike Content; Network Traffic Associated with Cobalt Strike Download; Anomalous Increase in System Resource Usage Management (SRUM)).

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 19 artifacts could be generated by the Program Download Technique
Technique Observers	IT Staff, IT Cybersecurity, OT Cybersecurity
Resources	Technique Detection References

3.6. REMOTE SYSTEM INFORMATION DISCOVERY TECHNIQUE (T0888) FOR DISCOVERY

Adversaries compromised the Windows AD server, causing Norsk Hydro to lose control of their entire IT infrastructure. Adversaries also used domain controller permissions to compromise all targeted files from the AD endpoints.²⁸ Associated Event IDs that could be found in the Windows Security Log are 4839 and 5136 to 5141.²⁹

Metasploit, Powershell Empire, Cobalt Strike, Mimikatz, and PsExec are all tools associated with the use of LockerGoga malware.^{30,31} Adversaries likely utilized PsExec to enable remote execution on the victim, creating copies of itself on various machines. They used Mimikatz and Metasploit to pull credentials from the AD server, PowerShell for remote code execution, and Cobalt Strike for port scanning and for remote service discovery.

IT Staff, IT Cybersecurity, and OT Cybersecurity personnel may have been able to observe anomalous traffic within the network.

A total of 12 observables were identified with the use of the [Remote System Information Discovery technique \(T0888\)](#). This technique is important for investigation because it serves as a vector to access additional systems, facilitating the spread of malware beyond the current target. This technique appears in the middle of the timeline and responding to it would likely prevent lateral movement within the environment. Terminating the chain of techniques at this point would prevent LockerGoga from spreading to other systems via lateral movement.

All 12 observables are assessed to be highly perceivable.

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of eight artifacts could be generated by the Remote System Discovery technique
Technique Observers	IT Staff, IT Cybersecurity, OT Cybersecurity
Resources	Technique Detection References

3.7. DATA FROM INFORMATION REPOSITORIES TECHNIQUE (T0811) FOR COLLECTION

LockerGoga gathers the victim's information and other identifiable information through the system, likely gathering administrative credentials, changing the user's login, and locking the user out of the system.³²

IT Staff and IT Cybersecurity personnel may have been able to observe adversaries capturing administrative credentials.

Two observables were identified with the use of the [Data from Information Repositories technique \(T0811\)](#). This technique is important for investigation because the victim could prevent credential harvesting and authorized user lock-out. This technique appears in the middle of the timeline and responding to it will limit adversaries' access to sensitive internal desktops. Terminating the chain of techniques at this point would limit extraction of proprietary information.

Both of the observables are assessed to be highly perceivable.

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 35 artifacts could be generated by the Data from Information Repositories technique
Technique Observers	IT Staff, IT Cybersecurity
Resources	Technique Detection References

3.8. DATA DESTRUCTION TECHNIQUE (T0809) FOR INHIBIT RESPONSE FUNCTION

Adversaries were able to destroy some backups and encrypt data within the Norsk Hydro network. After the victim opened the malicious attachment, the ransomware encrypted files via Crypto++ using RSA-4096 and AES-256 algorithms.³³ The LockerGoga malware encrypts documents, PDFs, spreadsheets, PowerPoint files, database files, videos, JavaScript, and Python files.³⁴ LockerGoga utilizes every CPU core and thread during the encryption process.³⁵

LockerGoga leaves a ransom note in a file named README_LOCKED.txt in the desktop folder, which is what Norsk Hydro employees found on a desktop workstation.³⁶

IT Staff, IT Cybersecurity, and OT Cybersecurity personnel may have been able to observe the file encryption.

A total of 41 observables were identified with the use of the [Data Destruction technique \(T0809\)](#). This technique is important for investigation because it encrypts victim files, rendering systems inoperable and effectively destroying data. This technique necessitates the unplanned restoration of data from backups and can result in unexpected expenses related to data recovery efforts. Appearing relatively late in the timeline, terminating the chain of techniques at this point would limit the destruction of data and resultant business interruptions.

Of the 41 observables associated with this technique, 38 are assessed to be highly perceivable (Anomalously Missing Files with extension bak, iso, doc, dot, docx, docb, dotx, wkb, xlm, xml, xls, xlsx, xlt, xltx, xlw, ppt, pps, pot, ppsx, pptx, posx, potx, sldx, pdf, db, sql, cs, ts, js, py); Hosts in Domain Anomalously Crash; Hosts Throw Error Messages; Host System with Anomalously Slow Response Time; Anomalous Increase in System Resource Usage Management (SRUM); Anomalously High System CPU Utilization; Anomalous Creation of Text File; Anomalous Creation of README_LOCKED.txt File).

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 27 artifacts could be generated by the Data Destruction technique
Technique Observers	IT Staff, IT Cybersecurity, OT Cybersecurity
Resources	Technique Detection References

3.9. SERVICE STOP TECHNIQUE (T0881) FOR INHIBIT RESPONSE FUNCTION

An operator was the first employee to notice the ransomware note on a PC at the Magnor extrusion plant, one of Norsk Hydro's 160 sites. The operator found the note at approximately 10:45 PM on 18 March.³⁷ Early on 19 March, Norsk Hydro's Head of Enterprise Service Management made the executive decision to shut down 23,000 PCs and 3,000 servers across the network.³⁸ Employees pulled power and network cables from outlets one by one to shut the systems down.³⁹ To successfully communicate the emergency shutdown internally, the Norsk Hydro communication team used Office 365 to communicate via email.⁴⁰ Norsk Hydro continued to spread information by hanging up flyers, posting on websites, and communicating through their social media pages.⁴¹

IT Staff, IT Cybersecurity, OT Staff, OT Cybersecurity, Engineering, Support Staff, and Management personnel may have been able to observe Norsk Hydro's shutdown.

A total of ten observables were identified with the use of the [Inhibit Response Function technique \(T0881\)](#). Appearing late in the timeline, terminating the chain of techniques at this point would limit product and delivery services to Norsk Hydro customers.

All ten observables are assessed to be highly perceivable.

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 13 artifacts could be generated by the Service Stop technique
Technique Observers	IT Staff, IT Cybersecurity, OT Staff, OT Cybersecurity, Engineering, Support Staff, and Management
Resources	Technique Detection References

3.10. LOSS OF AVAILABILITY TECHNIQUE (T0826) FOR IMPACT

Adversaries deleted and encrypted Human-Machine Interface (HMI) Controls, workstations, and databases.⁴² Employees continued operations manually, tracking manufacturing and finances with pen and paper. Employees were only able to connect devices to personal hotspots to provide limited functionality.⁴³ To avoid further damage the company decided not to pay the ransom, and instead planned to restore data from cloud backups.⁴⁴

IT Staff, IT Cybersecurity, OT Staff, OT Cybersecurity, Engineering, Support Staff, and Management personnel may have been able to observe encrypted HMI controls, workstations, and databases.

A total of 14 observables were identified with the use of the [Loss of Availability technique \(T0826\)](#). This technique is important for investigation because it prevents owners and operators from delivering products or services. This technique also presents noticeable effects, resulting in unresponsive equipment and limited network functionality. Appearing late in the timeline, terminating the chain of techniques at this point would limit the ability of LockerGoga to encrypt proprietary information.

All 14 observables are assessed to be highly perceivable.

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of eight artifacts could be generated by the Loss of Availability technique
Technique Observers	IT Staff, IT Cybersecurity, OT Staff, OT Cybersecurity, Engineering, Support Staff, and Management
Resources	Technique Detection References

3.11. LOSS OF CONTROL TECHNIQUE (T0827) FOR IMPACT

Nearly 35,000 Norsk Hydro employees lost control of their workstations, networks, and servers on 19 March when the company shut down its network.⁴⁵

IT Staff, IT Cybersecurity, OT Staff, OT Cybersecurity, Engineering, Support Staff, and Management personnel may have been able to observe control loss over workstations, networks, and servers.

A total of five observables were identified with the use of the [Loss of Control technique \(T0827\)](#). This technique is important for investigation because it prevents owners and operators from issuing commands to equipment. This technique also presents noticeable effects, particularly unresponsive equipment. Appearing after the triggering event, terminating the chain of techniques at this point would not limit destruction or business impacts.

All five observables are assessed to be highly perceivable.

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 13 artifacts could be generated by the Loss of Control technique
Technique Observers	IT Staff, IT Cybersecurity, OT Staff, OT Cybersecurity, Engineering, Support Staff, and Management
Resources	Technique Detection References

3.12. LOSS OF PRODUCTIVITY AND REVENUE TECHNIQUE (T0828) FOR IMPACT

Nearly a month after the cyber attack, the majority of Norsk Hydro's 160 business segments were still operating manually. The company suffered financial losses of approximately \$71 million, primarily in revenues from its Extruded Solutions business division, which was the most impacted by the ransomware attack.⁴⁶

IT Staff, IT Cybersecurity, OT Staff, OT Cybersecurity, Engineering, Support Staff, and Management personnel may have been able to observe manual operations and revenue loss.

A total of two observables were identified with the use of the [Loss of Productivity and Revenue technique \(T0828\)](#). This technique is important for investigation because it involves a direct loss of revenue and productivity for the victim. Additionally, this technique may present an impact for the end users or consumers of products and services. Terminating the chain of techniques at this point would not limit destruction or business impacts.

Both observables are assessed to be highly perceivable.

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 5 artifacts could be generated by the Loss of Productivity and Revenue technique
Technique Observers	IT Staff, IT Cybersecurity, OT Staff, OT Cybersecurity, Engineering, Support Staff, and Management
Resources	Technique Detection References

3.13. LOSS OF VIEW TECHNIQUE (T0829) FOR IMPACT

With the executive decision to shut down 23,000 PCs and 3,000 servers within the network, employees had to resort to manual operations for nearly a month, as they were unable to connect to their production systems.⁴⁷

IT Staff, IT Cybersecurity, OT Staff, OT Cybersecurity, Engineering, Support Staff, and Management personnel may have been able to observe the loss of view to operational systems.

A total of three observables were identified with the use of the [Loss of View technique \(T0829\)](#). This technique is important for investigation as it prevents owners and operators from delivering products or services. This technique also presents noticeable effects, such as non-responsive equipment. This technique appears at the end of the timeline, beyond the point at which a defender could take action to disrupt the attack.⁴⁸

All three observables are assessed to be highly perceivable.

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of four artifacts could be generated by the Loss of View technique
Technique Observers	IT Staff, IT Cybersecurity, OT Staff, OT Cybersecurity, Engineering, Support Staff, and Management
Resources	Technique Detection References

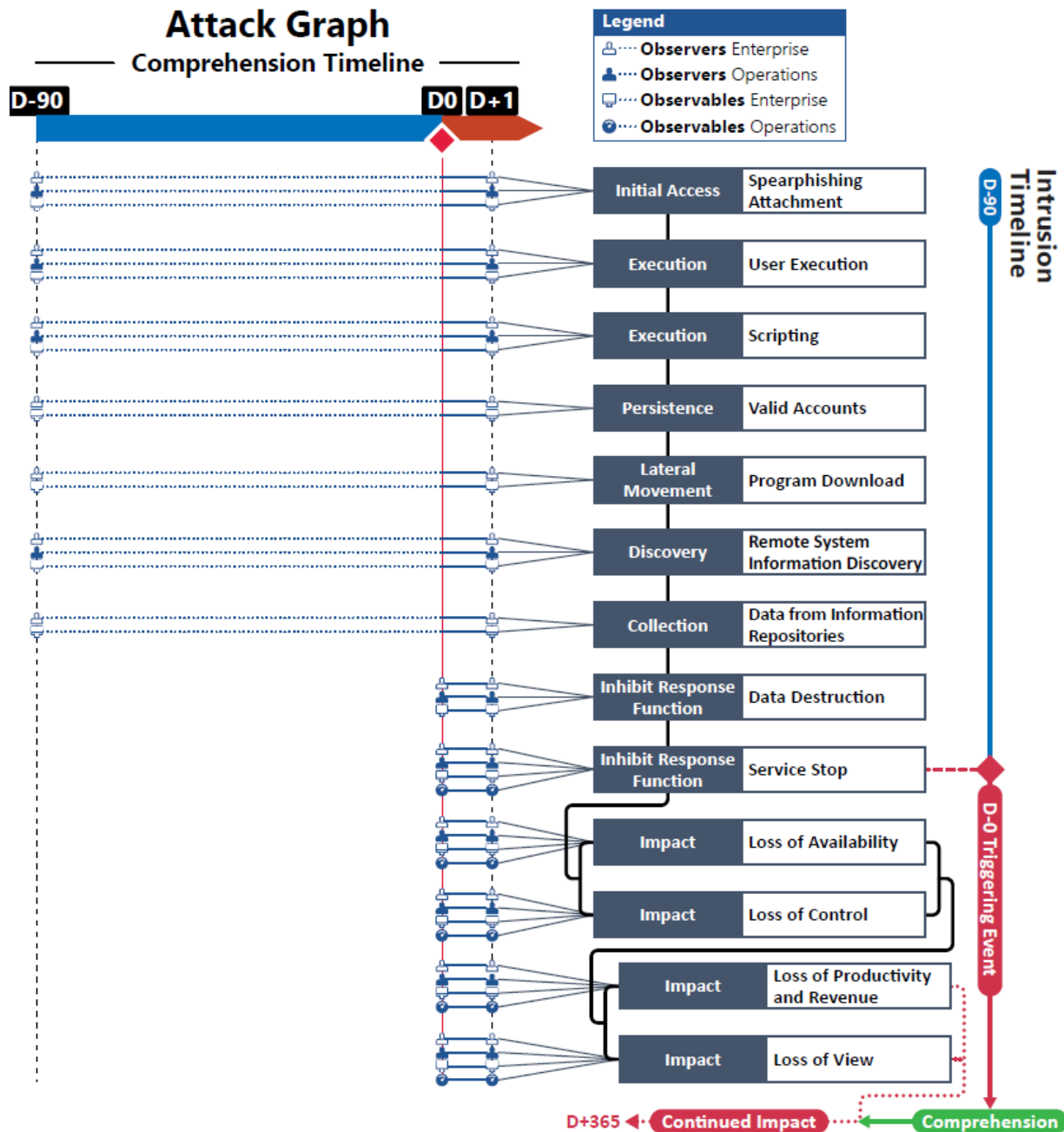


Figure 3. Attack Graph

APPENDIX A: OBSERVABLES LIBRARY

NOTE: Highly perceivable observables are highlighted in italics

Observables Associated with Spearphishing Attachment Technique (T0865)	
Observable 1	<i>Anomalous Email with Attachment</i>
Observable 2	<i>Email Attachment with Anomalous Embedded Scripts</i>
Observable 3	Anomalous SMTP Traffic Over TCP Port 25
Observable 4	<i>Email with Anomalous Word Document</i>
Observable 5	<i>Email with Anomalous Word Document with Object Linking and Embedding (OLE)</i>
Observable 6	<i>Email with Anomalous Excel File</i>
Observable 7	<i>Email with Anomalous Excel File with Object Linking and Embedding (OLE)</i>

Observables Associated with User Execution Technique (T0863)	
Observable 1	<i>Anomalous Access from External Host</i>
Observable 2	Anomalous User Collecting Credentials
Observable 3	Anomalous Remote Process Execution
Observable 4	<i>Anomalous Remote Process Execution Using Renamed PsExec</i>
Observable 5	Creation of Anomalous Executable Files
Observable 6	<i>Creation of Anomalous Executable Files in %TEMP%\svc{random}.{random number}.exe</i>
Observable 7	Anomalous File Execution
Observable 8	<i>Anomalous File Execution from %TEMP%\svc{random}.{random number}.exe-{random} -{random} {random}</i>
Observable 9	<i>Anomalous File Execution from %TEMP%\tgytutrc{4 Random Numbers}.exe</i>
Observable 10	<i>Anomalous Process Spawned from PsExec with the -m Parameter</i>
Observable 11	<i>Anomalous Process Spawned from Anomalous Binary Renamed from PsExec with the -m Parameter</i>
Observable 12	File Creation Windows Event Log (Sysmon Event ID: 11)
Observable 13	<i>A New Process Has Been Created Windows Event Log (Windows Event ID 4688)</i>

Observables Associated with Scripting Technique (T0853)	
Observable 1	Anomalous Traffic over TCP port 135
Observable 2	<i>Anomalous Traffic over TCP port 135 Associated with PsExec</i>
Observable 3	<i>Anomalous Traffic over TCP port 44</i>
Observable 4	<i>Anomalous Traffic over TCP port 44 Associated with PsExec</i>

Observables Associated with Scripting Technique (T0853)	
Observable 5	Anomalous Renamed PsExec Executable Downloaded to Internal Host
Observable 6	<i>A User Account was Successfully Logged on (Windows Event ID 4624)</i>
Observable 7	<i>A User Account Failed to Log on (Windows Event ID 4625)</i>
Observable 8	<i>A Service was Installed in the System (Windows Event ID 4697)</i>
Observable 9	<i>A New Service was Installed in the System (Windows Event ID 7045)</i>
Observable 10	Anomalous Execution of Command
Observable 11	Anomalous Execution of "task kill" Command
Observable 12	<i>Anomalous Processes Killed</i>
Observable 13	<i>Anomalous Anti-Virus Process Killed</i>
Observable 14	<i>Anomalous Execution of Script</i>
Observable 15	<i>Anomalous Execution of Logoff.exe Script</i>
Observable 16	<i>Anomalous Modification of Registry Key</i>
Observable 17	<i>Anomalous Modification of Registry Key: (HKEY_CURRENT_USER\SOFTWARE\Microsoft\RestartManager\Session00{01-20})</i>
Observable 18	Anomalous Creation of File
Observable 19	<i>Anomalous Creation of File Named README_LOCKED.txt</i>
Observable 20	Anomalous TCP Traffic Over Port 445
Observable 21	<i>Anomalous TCP Traffic Over Port 445 Associated with Mimikatz</i>
Observable 22	<i>An Account was Successfully Logged on Windows Log Event (Windows Event ID 4624)</i>
Observable 23	<i>Special Privileges Assigned to New Logon Windows Log Event (Windows Event ID 4672)</i>
Observable 24	Anomalous Executable Download
Observable 25	Anomalous mimikatz.exe Executable Download

Observables Associated with Valid Accounts Technique (T0859)	
Observable 1	<i>Anomalous Interactive Successful Logons (Windows Event ID 4624 Type 3 or 10)</i>
Observable 2	Anomalous Access to Administrator Credential Hashes
Observable 3	Anomalous System Logon Event Timestamp
Observable 4	<i>An Account was Successfully Logged on Windows Log Event (Windows Event ID 4624)</i>
Observable 5	<i>Special Privileges Assigned to New Logon Windows Log Event (Windows Event ID 4672)</i>

Observables Associated with Program Download (T0843)	
Observable 1	<i>Anomalous Application Event Log Entries</i>
Observable 2	<i>Application Event Log Entries Associated with Anomalous Executable</i>
Observable 3	<i>Anomalous File Installed</i>
Observable 4	<i>Anomalous File Installed Associated with LockerGoga</i>
Observable 5	<i>Anomalous Network Traffic Content</i>
Observable 6	<i>Network Traffic Associated with Anomalous Metasploit Content</i>
Observable 7	<i>Anomalous Network GET Request to an External Source</i>
Observable 8	<i>Network Traffic Associated with Anomalous Executable Download</i>
Observable 9	<i>Network Traffic Associated with Metasploit Download</i>
Observable 10	<i>Network Traffic Associated with Anomalous Mimikatz Content</i>
Observable 11	<i>Anomalous Network GET Request to an External Source</i>
Observable 12	<i>Network Traffic Associated with Anomalous Executable Download</i>
Observable 13	<i>Network Traffic Associated with Mimikatz Download</i>
Observable 14	<i>Network Traffic Associated with Anomalous Cobalt Strike Content</i>
Observable 15	<i>Anomalous Network GET Request to an External Source</i>
Observable 16	<i>Network Traffic Associated with Anomalous Executable Download</i>
Observable 17	<i>Network Traffic Associated with Cobalt Strike Download</i>
Observable 18	<i>Anomalous Increase in System Resource Usage Management (SRUM)</i>

Observables Associated with Remote System Information Discovery Technique (T0888)	
Observable 1	<i>Anomalous Internal Network Communication to Windows AD Domain Controller</i>
Observable 2	<i>Anomalous User Accessing Domain Controller</i>
Observable 3	<i>Anomalous User Attempted to Validate Credentials for An Account (Windows 4776)</i>
Observable 4	<i>A Domain Policy Was Changed to Domain Controller (Windows Event ID 4739)</i>
Observable 5	<i>Network Traffic Associated with Metasploit Download</i>
Observable 6	<i>Network Traffic Associated with Mimikatz Download</i>
Observable 7	<i>Anomalous Network Scanning Traffic Associated with Cobalt Strike Download</i>
Observable 8	<i>Anomalous Code Execution Across Internal Domain</i>
Observable 9	<i>Anomalous Code Execution Across Internal Domain Associated with PowerShell</i>
Observable 10	<i>Anomalous Traffic over TCP port 135 Associated with PsExec from Domain Controller to Host on Internal Domain</i>
Observable 11	<i>Anomalous Traffic over TCP port 44 Associated with PsExec from Domain Controller to Host on Internal Domain</i>
Observable 12	<i>A Directory Service Object Was Modified Event (Windows Event ID 5136)</i>

Observables Associated with Data from Information Repositories Technique (T0811)

Observable 1	<i>Anomalous Account Logon Information Changed (Windows Event ID 5136)</i>
Observable 2	<i>Anomalous Increase in User Account Logon Failures (Windows Event ID 4625)</i>

Observables Associated with Data Destruction Technique (T0809)

Observable 1	<i>Anomalously Missing Backup Files .bak</i>
Observable 2	<i>Anomalously Missing Image Files .iso</i>
Observable 3	<i>Anomalously Missing Files</i>
Observable 4	<i>Anomalously Missing Files with Extension (.doc)</i>
Observable 5	<i>Anomalously Missing Files with Extension (.dot)</i>
Observable 6	<i>Anomalously Missing Files with Extension (.docx)</i>
Observable 7	<i>Anomalously Missing Files with Extension (.docb)</i>
Observable 8	<i>Anomalously Missing Files with Extension (.dotx)</i>
Observable 9	<i>Anomalously Missing Files with Extension (.wkb)</i>
Observable 10	<i>Anomalously Missing Files with Extension (.xlm)</i>
Observable 11	<i>Anomalously Missing Files with Extension (.xml)</i>
Observable 12	<i>Anomalously Missing Files with Extension (.xls)</i>
Observable 13	<i>Anomalously Missing Files with Extension (.xlsx)</i>
Observable 14	<i>Anomalously Missing Files with Extension (.xlt)</i>
Observable 15	<i>Anomalously Missing Files with Extension (.xltx)</i>
Observable 16	<i>Anomalously Missing Files with Extension (.xlw)</i>
Observable 17	<i>Anomalously Missing Files with Extension (.ppt)</i>
Observable 18	<i>Anomalously Missing Files with Extension (.pps)</i>
Observable 19	<i>Anomalously Missing Files with Extension (.pot)</i>
Observable 20	<i>Anomalously Missing Files with Extension (.ppsx)</i>
Observable 21	<i>Anomalously Missing Files with Extension (.pptx)</i>
Observable 22	<i>Anomalously Missing Files with Extension (.posx)</i>
Observable 23	<i>Anomalously Missing Files with Extension (.potx)</i>
Observable 24	<i>Anomalously Missing Files with Extension (.sldx)</i>
Observable 25	<i>Anomalously Missing Files with Extension (.pdf)</i>
Observable 26	<i>Anomalously Missing Files with Extension (.db)</i>
Observable 27	<i>Anomalously Missing Files with Extension (.sql)</i>
Observable 28	<i>Anomalously Missing Files with Extension (.cs)</i>
Observable 29	<i>Anomalously Missing Files with Extension (.ts)</i>

Observables Associated with Data Destruction Technique (T0809)	
Observable 30	Anomalously Missing Files with Extension (.js)
Observable 31	Anomalously Missing Files with Extension (.py)
Observable 32	Hosts in Domain Anomalously Crash
Observable 33	Hosts Throw Error Messages
Observable 34	Host System with Anomalously Slow Response Time
Observable 35	Anomalous Increase in System Resource Usage Management (SRUM)
Observable 36	Anomalously High System CPU Utilization
Observable 37	Anomalous Call to Encryption Library
Observable 38	Anomalous Call to Crypto++ Encryption Library
Observable 39	Anomalous Call to RSA-2096 and AES-256 Encryption Libraries
Observable 40	Anomalous Creation of Text File
Observable 41	Anomalous Creation of README_LOCKED.txt File

Observables Associated with Service Stop Technique (T0881)	
Observable 1	Host Systems Shut Down for Extended Period of Time
Observable 2	23,000 Host Systems Shut Down for Extended Period of Time (30 days)
Observable 3	Network Servers Shut Down for Extended Period of Time
Observable 4	3,000 Network Servers Shut Down for Extended Period of Time (30 days)
Observable 5	Information Technology Processes Anomalously Fail
Observable 6	Operational Technology Processes Anomalously Fail
Observable 7	Information Technology Applications Anomalously Fail
Observable 8	Operational Technology Applications Anomalously Fail
Observable 9	Host Shutdown Event (Windows Event ID 4609)
Observable 10	Specific Windows Service Stopped (Windows Event ID 7040)

Observables Associated with Loss of Availability Technique (T0826)	
Observable 1	HMI Controls Unavailable
Observable 2	HMI Controls Unavailable for 30 days
Observable 3	Network Communication Unavailable
Observable 4	Network Communication Unavailable for 30 days
Observable 5	Host Applications Failing to Run
Observable 6	Server Applications Failing to Run
Observable 7	Host Applications Failing to Run for 30 days
Observable 8	Server Applications Failing to Run for 30 days

Observables Associated with Loss of Availability Technique (T0826)

Observable 9	<i>Inability to Utilize Services Within the IT Network</i>
Observable 10	<i>Inability to Utilize Services Within the IT Network for 30 days</i>
Observable 11	<i>Inability to Access Systems Within the OT Network</i>
Observable 12	<i>Inability to Access Systems Within the OT Network for 30 days</i>
Observable 13	<i>Encrypted System Files</i>
Observable 14	<i>Failed Attempt to Start Normal Functions</i>

Observables Associated with Loss of Control Technique (T0827)

Observable 1	<i>Emergency Shutdown Initiated for 30 days</i>
Observable 2	<i>Manual Operations for 30 days</i>
Observable 3	<i>Systems Remotely Inaccessible/Not Responsive for 30 days</i>
Observable 4	<i>Host Shutdown Event (Windows Event ID 4609)</i>
Observable 5	<i>Specific Windows Service Stopped (Windows Event ID 7040)</i>

Observables Associated with Loss of Productivity and Revenue Technique (T0828)

Observable 1	<i>Network Shut Down for Approximately Three Weeks</i>
Observable 2	<i>Approximately \$71 Million Lost in Revenue</i>

Observables Associated with Loss of View Technique (T0829)

Observable 1	<i>Loss of View of 23,000 PCs</i>
Observable 2	<i>Loss of View of 3,000 Servers</i>
Observable 3	<i>Loss of View of OT Operations</i>

APPENDIX B: ARTIFACTS LIBRARY

Artifacts Associated with Spearphishing Attachment Technique (T0865)	
Artifact 1	Email .ost File
Artifact 2	Mismatch MIME and Attachment File Extension
Artifact 3	Email Sender Address
Artifact 4	Email Message
Artifact 5	Email Receiver
Artifact 6	Email Receiver Name
Artifact 7	Email Receiver Domain
Artifact 8	Email Receiver Address
Artifact 9	Enable Macros Pop-Up
Artifact 10	Email Application Log File
Artifact 11	Email Unified Audit Log File
Artifact 12	Email Service Name
Artifact 13	Suspicious Email Message Content
Artifact 14	Email Sender Domain
Artifact 15	Email .pst File
Artifact 16	Email Sender IP Address
Artifact 17	Simple Mail Transfer Protocol SMTP Traffic
Artifact 18	Mail Transfer Agent Logs
Artifact 19	Email Parent Process
Artifact 20	Mail Transfer Agent Logs
Artifact 21	Email Domain Name System DNS Traffic
Artifact 22	Email Domain Name System DNS Event
Artifact 23	File Attachment Warning Prompt
Artifact 24	Email Timestamp
Artifact 25	Email Attachment
Artifact 26	Email Attachment File Type
Artifact 27	Email Header
Artifact 28	Email Sender Name
Artifact 29	Operating System Service Creation

Artifacts Associated with User Execution Technique (T0863)	
Artifact 1	Command Execution
Artifact 2	Service Termination

Artifacts Associated with User Execution Technique (T0863)	
Artifact 3	File Changes
Artifact 4	Increased ICMP Traffic (Network Scanning)
Artifact 5	Network Traffic Changes
Artifact 6	Application Installation
Artifact 7	Network Connection Creation
Artifact 8	Application Log Content
Artifact 9	User Account Modification
Artifact 10	File Creation
Artifact 11	Process Creation
Artifact 12	System Log
Artifact 13	Process Termination
Artifact 14	File Execution
Artifact 15	Prefetch Files
Artifact 16	Registry Modification
Artifact 17	File Modifications
Artifact 18	File Renaming
Artifact 19	System Patches Installed
Artifact 20	Files Opening
Artifact 21	File Signature Validation
Artifact 22	Installers Created
Artifact 23	Application Log

Artifacts Associated with Scripting Technique (T0853)	
Artifact 1	Startup Menu Modification
Artifact 2	OS Service Installation
Artifact 3	Registry Modifications
Artifact 4	Network Services Created
Artifact 5	External Network Connections
Artifact 6	Prefetch Files Created
Artifact 7	Executable Files
Artifact 8	System Processes Created
Artifact 9	OS Timeline Event
Artifact 10	System Event Log Creation
Artifact 11	Files Dropped into Directory

Artifacts Associated with Scripting Technique (T0853)

Artifact 12	Windows API Event Log
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Artifacts Associated with Valid Accounts Techniques (T0859)

Artifact 1	Logon Session Creation
Artifact 2	User Account Creation
Artifact 3	Logon Type Entry
Artifact 4	Logon Timestamp
Artifact 5	Failed Logons Event
Artifact 6	Successful Logon Event
Artifact 7	System Logs
Artifact 8	Default Credential Use
Artifact 9	Authentication Creation
Artifact 10	Prefetch Files Created After Execution
Artifact 11	Logons
Artifact 12	Application Log
Artifact 13	Domain Permission Requests
Artifact 14	Permission Elevation Requests
Artifact 15	Application Use Times
Artifact 16	Configuration Changes

Artifacts Associated with Program Download Technique (T0843)

Artifact 1	Controller State Change
Artifact 2	Controller Connection to External Website
Artifact 3	Controller In Stop State
Artifact 4	Controller Connected to External Networks
Artifact 5	Network Traffic Creation
Artifact 6	Network Metadata
Artifact 7	External IP Address
Artifact 8	Controller Network Connections via Management Protocol
Artifact 9	Operational Process Shutdown
Artifact 10	External Domain Connection
Artifact 11	Operational Process Restart
Artifact 12	Controller Application Log Type
Artifact 13	Supervisory Workstation Program Download Popup

Artifacts Associated with Program Download Technique (T0843)

Artifact 14	Controller Application Log Event
Artifact 15	Device Alarm
Artifact 16	Device Alert
Artifact 17	Operational Database Data Modification
Artifact 18	Controller Application Log Timestamp
Artifact 19	Controller In Program State

Artifacts Associated with Remote System Information Discovery Technique (T0888)

Artifact 1	Unexpected Recon Associated Library Calls
Artifact 2	Unexpected Standard Protocol Usage
Artifact 3	Unexpected Recon Associated Command Line Options (Ping Sweep, netstat, etc.)
Artifact 4	Unexpected Recon Associated Child Processes (Ping Sweep, netstat, etc.)
Artifact 5	Exfiltration of Host, Network, and/or System Architecture or Configuration Data
Artifact 6	Compromise and Exfiltration of Data from Asset Information Datastores or Applications
Artifact 7	Unexpected Industrial Protocol Usage
Artifact 8	Unexpected Industrial Application Usage

Artifacts Associated with Data from Information Repositories Technique (T0811)

Artifact 1	SFTP Traffic Port
Artifact 2	Share Drive Access
Artifact 3	Operational Database Logons
Artifact 4	Engineering Workstation Application Log
Artifact 5	HTTP Traffic Port
Artifact 6	HTTPS Traffic Port
Artifact 7	FTPS Traffic Port
Artifact 8	File Access
Artifact 9	Telnet Traffic Port
Artifact 10	File Modification
Artifact 11	FTP Traffic Port
Artifact 12	VNC Traffic Port
Artifact 13	RDP Traffic Port
Artifact 14	Authentication Success
Artifact 15	Authentication Attempts

Artifacts Associated with Data from Information Repositories Technique (T0811)	
Artifact 16	MSSQL Traffic
Artifact 17	Traffic Timestamps
Artifact 18	SMB Traffic
Artifact 19	Project File Modification
Artifact 20	Data Bytes Sent
Artifact 21	User Session Creation
Artifact 22	Application Logon
Artifact 23	TDS Port
Artifact 24	Operational Database Data Modification
Artifact 25	Design Documentation Manipulation
Artifact 26	Authentication Failure
Artifact 27	Personnel List Files Accessed
Artifact 28	Jump Host Credentials Accessed
Artifact 29	Vendor Documentation Accessed
Artifact 30	Remote Procedure Calls
Artifact 31	Recent Search List
Artifact 32	MRU List Change
Artifact 33	Design Documentation Access
Artifact 34	Database Request
Artifact 35	SSH Traffic Port

Artifacts Associated with Data Destruction Technique (T0809)	
Artifact 1	Command Line Arguments
Artifact 2	Files Moved to Recycle Bin
Artifact 3	Missing Files
Artifact 4	Host System Reboot Failure
Artifact 5	Process Logic Failure
Artifact 6	Event Log Creation
Artifact 7	System Call
Artifact 8	System Application Interruption
Artifact 9	Device Failure
Artifact 10	Recovery Attempt Failure
Artifact 11	TFTP Port
Artifact 12	SFTP Port

Artifacts Associated with Data Destruction Technique (T0809)	
Artifact 13	Memory Corruption
Artifact 14	Use of File Transfer Protocols
Artifact 15	SCP Port
Artifact 16	File Encryptions
Artifact 17	Non-Native Files
Artifact 18	External Network Connections
Artifact 19	Transient Device Connections
Artifact 20	Program Execution
Artifact 21	Telnet Port
Artifact 22	FTPS Port
Artifact 23	HTTP Port
Artifact 24	HTTPS Port
Artifact 25	Local Network Connections
Artifact 26	FTP Port
Artifact 27	SMB Port

Artifacts Associated with Service Stop Technique (T0881)	
Artifact 1	Internal System Logs
Artifact 2	Alarm Event
Artifact 3	OS API Call
Artifact 4	Application Error Messages
Artifact 5	Process Error Messages
Artifact 6	Application Service Stop
Artifact 7	Registry Change HKLM\SYSTEM\CURRENTCONTROLSET\SERVICES
Artifact 8	OS Service Crash
Artifact 9	System Event Logs
Artifact 10	Application Event Logs
Artifact 11	System Resource Usage Manager Application Usage Change
Artifact 12	Command Line System Argument
Artifact 13	Process Failure

Artifacts Associated with Loss of Availability Technique (T0826)	
Artifact 1	Process Failure Due to Loss of Required Network or System Dependency
Artifact 2	Unexplained Loss of User Data

Artifacts Associated with Loss of Availability Technique (T0826)	
Artifact 3	Changes In Network Routing or Usage of Redundant Control System Network Connection Due to Failed Network Path
Artifact 4	Significant Reduction or Increase in Network Traffic Due to Malware Propagation of Disappearance of Services
Artifact 5	Significant Logged Usage of Native Crypto Functions or Presence of Import of Crypto Functions in Binaries
Artifact 6	Operator or User Discovery of Encrypted or Inoperable Systems
Artifact 7	File System Modification Artifacts Might Be Associated with The Loss of Availability Might Be Present on Disk
Artifact 8	Unexplained Loss of Application Data

Artifacts Associated with Loss of Control Technique (T0827)	
Artifact 1	Failed Input Commands
Artifact 2	Repeated Maintenance Reports
Artifact 3	Process Failure
Artifact 4	Unresponsive I/O Conditions
Artifact 5	Network Connection Loss
Artifact 6	Process Environment Changes
Artifact 7	Runaway Conditions
Artifact 8	Service Request Increases
Artifact 9	Set Point Failure
Artifact 10	Configuration Change
Artifact 11	Machine State Change
Artifact 12	Process Alarms
Artifact 13	Device Failure

Artifacts Associated with Loss of Productivity and Revenue Technique (T0828)	
Artifact 1	Loss of Confidence in a Safety System Due to Unreliability Might Result in a Risk Management Driven Shutdown of a Plant
Artifact 2	Wormable or Other Highly Propagating Malware Might Result in the Shutdown of a Plant to Prevent Ransomware or Other Destructive Attacks
Artifact 3	Extortion Attempts Might Lead to Reduced Operations Due to Potential Presence of Malicious Attackers
Artifact 4	Loss of Control of Critical Systems Due to Ransomware or Loss of Confidence Might Lead to a Degraded Productivity or Revenue Operating State
Artifact 5	File System Modification Artifacts Might Be Associated with The Loss of Productivity and Revenue Attack Might Be Present on Disk

Artifacts Associated with Loss of View Technique (T0829)	
Artifact 1	Blocking Communications Paths and Channels (Comms, Network Infrastructure, Host Packet Routing) Might Occur via an OS or Device Modification
Artifact 2	Masked Reconfiguration of Alarm Thresholds or Other Configuration Settings Might Result in Unexpected Lack of Situational Awareness
Artifact 3	File System Modification Artifacts Might Be Associated with The Loss of View Attack Might Be Present on Disk
Artifact 4	Application Logic Hooks or Modifications Might Prevent Proper Reporting to/from an Industrial Application

APPENDIX C: OBSERVERS

This is a collection of standardized potential observers that work in operational technology organizations. It has been slightly modified by the CyOTE team from the [Job Role Groupings listed in the SANS ICS Job Role to Competency Level Poster](#) to communicate the categories of potential observers during cyber events.

Engineering  • Process Engineer • Electrical, Controls, and Mechanical Engineer • Project Engineer • Systems and Reliability Engineer • OT Developer • PLC Programmer • Emergency Operations Manager • Plant Networking • Control/Instrumentation Specialist • Protection and Controls • Field Engineer • System Integrator	Support Staff  • Remote Maintenance & Technical Support • Contractors (engineering) • IT and Physical Security Contractor • Procurement Specialist • Legal • Contracting Engineer • Insurance • Supply-chain Participant • Inventory Management/Lifecycle Management • Physical Security Specialist
Operations Technology (OT) Staff  • Operator • Site Security POC • Technical Specialists (electrical/mechanical/chemical) • ICS/SCADA Programmer	Information Technology (IT) Cybersecurity  • ICS Security Analyst • Security Engineering and Architect • Security Operations • Security Response and Forensics • Security Management (CSO) • Audit Specialist
Operational Technology (OT) Cybersecurity  • OT Security • ICS/SCADA Security	• Security Tester
Management  • Plant Manager • Risk/Safety Manager • Business Unit Management • C-level Management	Information Technology (IT) Staff  • Networking and Infrastructure • Host Administrator • Database Administrator • Application Development • ERP/MES Administrator • IT Management

REFERENCES

- ¹ [Norsk Hydro | “Power and market operations” | <https://www.hydro.com/en-US/energy/power-and-market-operations/> | 26 July 2021 | Accessed 18 May 2022 | The source is publicly available information and does not contain classification markings]
- ² [YouTube | Insignia – Crisis Management Specialists | Jonathan Hemus and Inger Sethov | “Webinar: How Norsk Hydro emerged from a cyber attack with its reputation enhanced” | <https://www.youtube.com/watch?v=C6MDz-AgQuE&t=12s> | 22 April 2021 | Accessed 18 May 2022 | The source is publicly available information and does not contain classification markings]
- ³ [YouTube | Norsk Hydro | “The cyber attack rescue operation in Hydro Toulouse” | <https://www.youtube.com/watch?v=o6eEN0mUakM> | 16 April 2021 | Accessed 18 May 2022 | The source is publicly available information and does not contain classification markings]
- ⁴ [Microsoft | Bill Briggs | “Hackers hit Norsk Hydro with ransomware. The company responded with transparency” | <https://news.microsoft.com/transform/hackers-hit-norsk-hydro-ransomware-company-responded-transparency/> | 16 December 2019 | Accessed 18 May 2022 | The source is publicly available information and does not contain classification markings]
- ⁵ [Microsoft | Bill Briggs | “Hackers hit Norsk Hydro with ransomware. The company responded with transparency” | <https://news.microsoft.com/transform/hackers-hit-norsk-hydro-ransomware-company-responded-transparency/> | 16 December 2019 | Accessed 18 May 2022 | The source is publicly available information and does not contain classification markings]
- ⁶ [YouTube | Norsk Hydro | “Cyber attack on Hydro Magnor” | <https://www.youtube.com/watch?v=S-ZlVuM0we0> | 28 October 2019 | Accessed 18 May 2022 | The source is publicly available information and does not contain classification markings]
- ⁷ [DoublePulsar | Kevin Beaumont | “How Lockergoga took down Hydro – ransomware used in targeted attacks aimed at big business” | <https://doublepulsar.com/how-lockergoga-took-down-hydro-ransomware-used-in-targeted-attacks-aimed-at-big-business-c666551f5880> | 21 Mar 2019 | Accessed 18 May 2022 | The source is publicly available information and does not contain classification markings]
- ⁸ [YouTube | Insignia – Crisis Management Specialists | Jonathan Hemus and Inger Sethov | “Webinar: How Norsk Hydro emerged from a cyber attack with its reputation enhanced” | <https://www.youtube.com/watch?v=C6MDz-AgQuE&t=12s> | 22 April 2021 | Accessed 18 May 2022 | The source is publicly available information and does not contain classification markings]
- ⁹ [YouTube | Norsk Hydro | “Why Hydro chose to be transparent during cyber-attack” | <https://www.youtube.com/watch?v=C6MDz-AgQuE&t=12s> | 28 October 2019 | Accessed 18 May 2022 | The source is publicly available information and does not contain classification markings]
- ¹⁰ [YouTube | Norsk Hydro | “The cyber attack rescue operation in Hydro Toulouse” | <https://www.youtube.com/watch?v=o6eEN0mUakM> | 16 April 2021 | Accessed 18 May 2022 | The source is publicly available information and does not contain classification markings]
- ¹¹ [ZDNet | Catalin Cimpanu | “Norsk Hydro ransomware incident losses reach \$40 million after one week” | <https://www.zdnet.com/article/norsk-hydro-ransomware-incident-losses-reach-40-million-after-one-week/> | 26 March 2019 | Accessed 18 May 2022 | The source is publicly available information and does not contain classification markings]
- ¹² [Microsoft | Bill Briggs | “Hackers hit Norsk Hydro with ransomware. The company responded with transparency” | <https://news.microsoft.com/transform/hackers-hit-norsk-hydro-ransomware-company-responded-transparency/> | 16 December 2019 | Accessed 18 May 2022 | The source is publicly available information and does not contain classification markings]
- ¹³ [Microsoft | Bill Briggs | “Hackers hit Norsk Hydro with ransomware. The company responded with transparency” | <https://news.microsoft.com/transform/hackers-hit-norsk-hydro-ransomware-company-responded-transparency/> | 16 December 2019 | Accessed 18 May 2022 | The source is publicly available information and does not contain classification markings]

-
- ¹⁴ [TrendMicro | “What You Need to Know About the LockerGoga Ransomware” | <https://www.trendmicro.com/vinfo/us/security/news/cyber-attacks/what-you-need-to-know-about-the-lockergoga-ransomware> | 20 March 2019 | Accessed 18 May 2022 | The source is publicly available information and does not contain classification markings]
- ¹⁵ [Dragos | Joe Slowik | “Spyware Stealer Locker Wiper: LockerGoga Revisited” | <https://www.dragos.com/resource/spyware-stealer-locker-wiper-lockergoga-revisited/> | 16 March 2020 | Accessed 18 May 2022 | The source is publicly available information and does not contain classification markings]
- ¹⁶ [WIRED| Andy Greenburg | “Meet LockerGoga, the Ransomware Crippling Industrial Firms” | <https://www.wired.com/story/lockergoga-ransomware-crippling-industrial-firms/> | 25 March 2019 | Accessed 18 May 2022 | The source is publicly available information and does not contain classification markings]
- ¹⁷ [TrendMicro | “What You Need to Know About the LockerGoga Ransomware” | <https://www.trendmicro.com/vinfo/us/security/news/cyber-attacks/what-you-need-to-know-about-the-lockergoga-ransomware> | 20 March 2019 | Accessed 18 May 2022 | The source is publicly available information and does not contain classification markings]
- ¹⁸ [Microsoft | Justin Hall, Andrea Bichsel, Nick Schonning, and others | “5136(S): A directory service object was modified.” | <https://docs.microsoft.com/en-us/windows/security/threat-protection/auditing/event-5136> | 15 December 2021 | Accessed 18 May 2022 | The source is publicly available information and does not contain classification markings]
- ¹⁹ [TrendMicro | “What You Need to Know About the LockerGoga Ransomware” | <https://www.trendmicro.com/vinfo/us/security/news/cyber-attacks/what-you-need-to-know-about-the-lockergoga-ransomware> | 20 March 2019 | Accessed 18 May 2022 | The source is publicly available information and does not contain classification markings]
- ²⁰ [YouTube | Norsk Hydro | “Cyber attack on Hydro Magnor” | <https://www.youtube.com/watch?v=S-ZlVuM0we0> | 28 October 2019 | Accessed 18 May 2022 | The source is publicly available information and does not contain classification markings]
- ²¹ [Microsoft | Bill Briggs | “Hackers hit Norsk Hydro with ransomware. The company responded with transparency” | <https://news.microsoft.com/transform/hackers-hit-norsk-hydro-ransomware-company-responded-transparency/> | 16 December 2019 | Accessed 18 May 2022 | The source is publicly available information and does not contain classification markings]
- ²² [Fastmarkets AMM | Andrea Hotter | “How the Norsk Hydro cyberattack unfolded” | <https://www.amm.com/Article/3890250/How-the-Norsk-Hydro-cyberattack-unfolded.html> | 22 August 2019 | Accessed 18 May 2022 | The source is publicly available information and does not contain classification markings]
- ²³ [DoublePulsar | Kevin Beaumont | “How Lockergoga took down Hydro – ransomware used in targeted attacks aimed at big business” | <https://doublepulsar.com/how-lockergoga-took-down-hydro-ransomware-used-in-targeted-attacks-aimed-at-big-business-c666551f5880> | 21 Mar 2019 | Accessed 18 May 2022 | The source is publicly available information and does not contain classification markings]
- ²⁴ [McAfee | ATR Operational Intelligence Team, and Marc RiveroLopez | “LockerGoga Ransomware Family Used in Targeted Attacks” | <https://www.mcafee.com/blogs/other-blogs/mcafee-labs/lockergoga-ransomware-family-used-in-targeted-attacks/> | 29 April 2019 | Accessed 18 May 2022 | The source is publicly available information and does not contain classification markings]
- ²⁵ [WIRED| Andy Greenburg | “Meet LockerGoga, the Ransomware Crippling Industrial Firms” | <https://www.wired.com/story/lockergoga-ransomware-crippling-industrial-firms/> | 25 March 2019 | Accessed 18 May 2022 | The source is publicly available information and does not contain classification markings]
- ²⁶ [McAfee | ATR Operational Intelligence Team, and Marc RiveroLopez | “LockerGoga Ransomware Family Used in Targeted Attacks” | <https://www.mcafee.com/blogs/other-blogs/mcafee-labs/lockergoga-ransomware-family-used-in-targeted-attacks/> | 29 April 2019 | Accessed 18 May 2022 | The source is publicly available information and does not contain classification markings]

-
- ²⁷ [SciVerse ScienceDirect | Suvi Leppänen, Shohel Ahmed, and Robin Granqvist | “Cyber Security Incident Report – Norsk Hydro” | https://mycourses.aalto.fi/pluginfile.php/923542/mod_folder/content/0/Group%20CSS%20Norsk%20Hydro%202019.pdf | March 2019 | Accessed 18 May 2022 | The source is publicly available information and does not contain classification markings]
- ²⁸ [SciVerse ScienceDirect | Suvi Leppänen, Shohel Ahmed, and Robin Granqvist | “Cyber Security Incident Report – Norsk Hydro” | https://mycourses.aalto.fi/pluginfile.php/923542/mod_folder/content/0/Group%20CSS%20Norsk%20Hydro%202019.pdf | March 2019 | Accessed 18 May 2022 | The source is publicly available information and does not contain classification markings]
- ²⁹ [Microsoft | Justin Hall, Andrea Bichsel, Nick Schonning, and others | “54624(S): An account was successfully logged on.” | <https://docs.microsoft.com/en-us/windows/security/threat-protection/auditing/event-4624> | 14 December 2021 | Accessed 18 May 2022 | The source is publicly available information and does not contain classification markings]
- ³⁰ [Dragos | Joe Slowik | “Spyware Stealer Locker Wiper: LockerGoga Revisited” | <https://www.dragos.com/resource/spyware-stealer-locker-wiper-lockergoga-revisited/> | 16 March 2020 | Accessed 18 May 2022 | The source is publicly available information and does not contain classification markings]
- ³¹ [Microsoft | Justin Hall, Andrea Bichsel, Nick Schonning, and others | “54624(S): An account was successfully logged on.” | <https://docs.microsoft.com/en-us/windows/security/threat-protection/auditing/event-4624> | 14 December 2021 | Accessed 18 May 2022 | The source is publicly available information and does not contain classification markings]
- ³² [TrendMicro | “What You Need to Know About the LockerGoga Ransomware” | <https://www.trendmicro.com/vinfo/us/security/news/cyber-attacks/what-you-need-to-know-about-the-lockergoga-ransomware> | 20 March 2019 | Accessed 18 May 2022 | The source is publicly available information and does not contain classification markings]
- ³³ [Fastmarkets AMM | Andrea Hotter | “How the Norsk Hydro cyberattack unfolded” | <https://www.amm.com/Article/3890250/How-the-Norsk-Hydro-cyberattack-unfolded.html> | 22 August 2019 | Accessed 18 May 2022 | The source is publicly available information and does not contain classification markings]
- ³⁴ [TrendMicro | “What You Need to Know About the LockerGoga Ransomware” | <https://www.trendmicro.com/vinfo/us/security/news/cyber-attacks/what-you-need-to-know-about-the-lockergoga-ransomware> | 20 March 2019 | Accessed 18 May 2022 | The source is publicly available information and does not contain classification markings]
- ³⁵ [SciVerse ScienceDirect | Suvi Leppänen, Shohel Ahmed, and Robin Granqvist | “Cyber Security Incident Report – Norsk Hydro” | https://mycourses.aalto.fi/pluginfile.php/923542/mod_folder/content/0/Group%20CSS%20Norsk%20Hydro%202019.pdf | March 2019 | Accessed 18 May 2022 | The source is publicly available information and does not contain classification markings]
- ³⁶ [TrendMicro | “What You Need to Know About the LockerGoga Ransomware” | <https://www.trendmicro.com/vinfo/us/security/news/cyber-attacks/what-you-need-to-know-about-the-lockergoga-ransomware> | 20 March 2019 | Accessed 18 May 2022 | The source is publicly available information and does not contain classification markings]
- ³⁷ [Microsoft | Bill Briggs | “Hackers hit Norsk Hydro with ransomware. The company responded with transparency” | <https://news.microsoft.com/transform/hackers-hit-norsk-hydro-ransomware-company-responded-transparency/> | 16 December 2019 | Accessed 18 May 2022 | The source is publicly available information and does not contain classification markings]
- ³⁸ [YouTube | Insignia – Crisis Management Specialists | Jonathan Hemus and Inger Sethov | “Webinar: How Norsk Hydro emerged from a cyber attack with its reputation enhanced” | <https://www.youtube.com/watch?v=C6MDz-AgQuE&t=12s> | 22 April 2021 | Accessed 18 May 2022 | The source is publicly available information and does not contain classification markings]

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- ³⁹ [SciVerse ScienceDirect | Suvi Leppänen, Shohel Ahmed, and Robin Granqvist | “Cyber Security Incident Report – Norsk Hydro”
| https://mycourses.aalto.fi/pluginfile.php/923542/mod_folder/content/0/Group%20CSS%20Norsk%20Hydro%202019.pdf | March 2019 | Accessed 18 May 2022 | The source is publicly available information and does not contain classification markings]
- ⁴⁰ [YouTube | Norsk Hydro | “Why Hydro chose to be transparent during cyber-attack”
| <https://www.youtube.com/watch?v=C6MDz-AgQuE&t=12s> | 28 October 2019 | Accessed 18 May 2022 | The source is publicly available information and does not contain classification markings]
- ⁴¹ [YouTube | Insignia – Crisis Management Specialists | Jonathan Hemus and Inger Sethov | “Webinar: How Norsk Hydro emerged from a cyber attack with its reputation enhanced”
| <https://www.youtube.com/watch?v=C6MDz-AgQuE&t=12s> | 22 April 2021 | Accessed 18 May 2022 | The source is publicly available information and does not contain classification markings]
- ⁴² [Dragos | Joe Slowik | “Spyware Stealer Locker Wiper: LockerGoga Revisited”
| <https://www.dragos.com/resource/spyware-stealer-locker-wiper-lockergoga-revisited/> | 16 March 2020 | Accessed 18 May 2022 | The source is publicly available information and does not contain classification markings]
- ⁴³ [YouTube | Insignia – Crisis Management Specialists | Jonathan Hemus and Inger Sethov | “Webinar: How Norsk Hydro emerged from a cyber attack with its reputation enhanced”
| <https://www.youtube.com/watch?v=C6MDz-AgQuE&t=12s> | 22 April 2021 | Accessed 18 May 2022 | The source is publicly available information and does not contain classification markings]
- ⁴⁴ [ZDNet | Catalin Cimpanu | “Norsk Hydro ransomware incident losses reach \$40 million after one week”
| <https://www.zdnet.com/article/norsk-hydro-ransomware-incident-losses-reach-40-million-after-one-week/> | 26 March 2019 | Accessed 18 May 2022 | The source is publicly available information and does not contain classification markings]
- ⁴⁵ [Microsoft | Bill Briggs | “Hackers hit Norsk Hydro with ransomware. The company responded with transparency” | <https://news.microsoft.com/transform/hackers-hit-norsk-hydro-ransomware-company-responded-transparency/> | 16 December 2019 | Accessed 18 May 2022 | The source is publicly available information and does not contain classification markings]
- ⁴⁶ [ZDNet | Catalin Cimpanu | “Norsk Hydro ransomware incident losses reach \$40 million after one week”
| <https://www.zdnet.com/article/norsk-hydro-ransomware-incident-losses-reach-40-million-after-one-week/> | 26 March 2019 | Accessed 18 May 2022 | The source is publicly available information and does not contain classification markings]
- ⁴⁷ [TrendMicro | “What You Need to Know About the LockerGoga Ransomware”
| <https://www.trendmicro.com/vinfo/us/security/news/cyber-attacks/what-you-need-to-know-about-the-lockergoga-ransomware> | 20 March 2019 | Accessed 18 May 2022 | The source is publicly available information and does not contain classification markings]
- ⁴⁸ [YouTube | Insignia – Crisis Management Specialists | Jonathan Hemus and Inger Sethov | “Webinar: How Norsk Hydro emerged from a cyber attack with its reputation enhanced”
| <https://www.youtube.com/watch?v=C6MDz-AgQuE&t=12s> | 22 April 2021 | Accessed 18 May 2022 | The source is publicly available information and does not contain classification markings]